Woojun Jung

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RESEARCH INTERESTS

Memory in AI: using mechanistic interpretability to study how models encode and use experience, and developing memory architectures and learning methods for reliable recall and knowledge revision.

Longitudinal User Modeling: uncertainty-aware modeling of user traits across time and contexts, and developing human-in-the-loop evaluations of adaptation and continuity in long-term human-AI relationships.

Multimodal Social Reasoning: integrating verbal and nonverbal evidence to interpret shared experiences and revise inferences about people's emotional states, intentions, and perspectives.

EDUCATION

Korea University *Mar 2025 – Feb 2027 (Expected)*
M.S. in Computer Science and Engineering
Advisor: Prof. Susik Yoon

Korea University *Mar 2022 – Feb 2025*
B.E. in Computer Science and Engineering

Korea University *Mar 2016 – Feb 2022*
B.F.A. in Art and Design *2 years of Military Service (Oct. 2017 – Sep. 2019)*

Relevant Coursework: *CS/AI* - Algorithms (A+), Computer Architecture (A), Deep Learning (A), Optimization Theory (A), Reinforcement Learning (A), Natural Language Processing (A), Data Science (A), Brain and Machine Cognition (A+); *Design/HCI* - Interaction Design (A+), Interface Design (A+), Design and Human Factors (A+), Design Methodology (A+).

PUBLICATIONS

[3] **Segment-driven Structural Induction and Semantic Alignment for Heterogeneous Tabular Representation**
Woojun Jung and Susik Yoon
ICML 2026 — [PMLR] [Code]

[2] **Can Structural Cues Save LLMs? Evaluating Language Models in Massive Document Streams**
Yukyung Lee, Yebin Lim*, Woojun Jung*, Wonjun Choi, Susik Yoon
KDD 2026 — [ACM DL] [Code] (*: equal contribution)

[1] **HAETAE: In-Domain Table Pretraining with Header Anchoring**
Woojun Jung and Susik Yoon
SIGIR 2025 (Short) — [ACM DL] [Code]

RESEARCH EXPERIENCE

DAIS Lab, Korea University *Aug 2024 – Present*
Research Intern (*Aug 2024 – Feb 2025*); Research Assistant (*Mar 2025 – Present*)

Uncertainty-aware User Modeling for Conversational Agents *May 2026 – Present*

- Defining long-horizon tasks for response personalization and agent action selection.
- Designing evaluation protocols to assess calibration of predicted probabilities that remembered preferences apply in new contexts, using subsequent user judgments.
- Exploring how to estimate uncertainty from learned representations and whether these estimates can improve response personalization and clarification decisions.

Benchmarking LLM Understanding of Evolving Facts in Document Streams **KDD 2026**

- Developed the Temporal QA task, one of three benchmark tasks alongside Topic Clustering and Summarization.
- Designed and implemented its evaluation protocol, benchmark dataset, and human-annotation guidelines.

Context-dependent Attribute Semantics in Tabular Data **ICML 2026; SIGIR 2025**

- Led two first-author projects on domain-specific tabular representation learning.
- Proposed header anchoring to regularize contextualized header embeddings toward context-independent anchors, stabilizing semantics across row contexts and schema variants.
- Developed a pretraining method that jointly learns schema and column contexts with entropy-routed objectives to preserve domain-coherent and instance-specific attribute semantics.

APPLIED RESEARCH COLLABORATION

Foundation Models for Enterprise Relational Data, KT (Korea Telecom) *2025 – 2026*

- Collaborated with KT on foundation-model methods for enterprise relational databases.
- Analyzed challenges in adapting tabular ML to organizational data with heterogeneous schemas.
- Prototyped a pretraining pipeline for schema-aware attribute representations for enterprise tables.
- Developed a schema-linking method that maps natural-language questions to relevant tables and columns.

SELECTED PROJECTS

Dr.Snap: AI-guided Dermatology Intake — [Link] *2024*

- Built a conversational intake prototype that elicits symptom details through adaptive follow-up questions informed by retrieved condition information and prior responses.
- Applied hierarchical task analysis to propose an integrated workflow connecting standalone AI guidance with clinic intake, positioning AI as pre-visit support.

Silverlining: Presence-aware Adaptive Lighting — [Link] *2020*

- Conducted interviews and observations of home lighting use, identifying manual adjustments during activity transitions and informing the design of a presence-responsive pendant-light concept.

Bianca: Full-Scale Interactive Refrigerator Prototype — [Link] *2020*

- Developed Arduino code and electronics and built a full-scale refrigerator prototype that maps handle-press locations to corresponding illuminated regions on the door.

TEACHING EXPERIENCE

Teaching Assistant, Data Science, Korea University *Spring 2025*

- Prepared and delivered tutorials on LaTeX and visualization with matplotlib for undergraduate students.

Teaching Assistant, Artificial Intelligence, Korea University *Fall 2025*

- Prepared and delivered a tutorial on multimodal representation learning with CLIP for undergraduate students.

AWARDS & SCHOLARSHIPS

Master's Fellowship (KRW 12,000,000), National Research Foundation of Korea (NRF) *2025–2026*
Second Place, Korea University AI Tech Day Poster Presentation *2025*
Third Place, Korea University Capstone Design Competition *2024*
Third Place, Samsung Design Competition *2019*

SKILLS

Programming & Frameworks: *Advanced*— Python, PyTorch, Transformers, pandas, NumPy, matplotlib, scikit-learn, Git, GitHub, LaTeX; *Intermediate*— C, CUDA, SQL; *Novice*— vLLM, SGLang;

Technical & Research Skills: *Advanced*— Applied Machine Learning, Natural Language Processing, Data Analysis; *Intermediate*— Bayesian Inference, Reinforcement Learning, Human-Computer Interaction;

Design & Prototyping: *Advanced*— HTML, CSS, Figma, Illustrator, Photoshop, Rhinoceros, Keyshot; *Intermediate*— JavaScript, React, Arduino, 3D printing;

Languages: *Native*—Korean, *Fluent*—English (TOEFL iBT 5.5/6.0)